

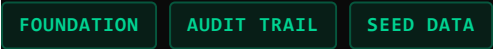
01 — EXECUTIVE SUMMARY

Matching velocity. Compliance certainty.

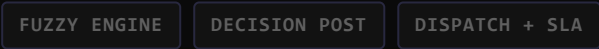
VeloIQ is TÜV Rheinland's automated standards pipeline — ingesting certification standards, fuzzy-matching them against customer certificates, capturing expert TCC impact assessments, and dispatching customer notifications with immutable audit trail enforcement.



PHASE A — LIVE



PHASE B-D — DEFERRED

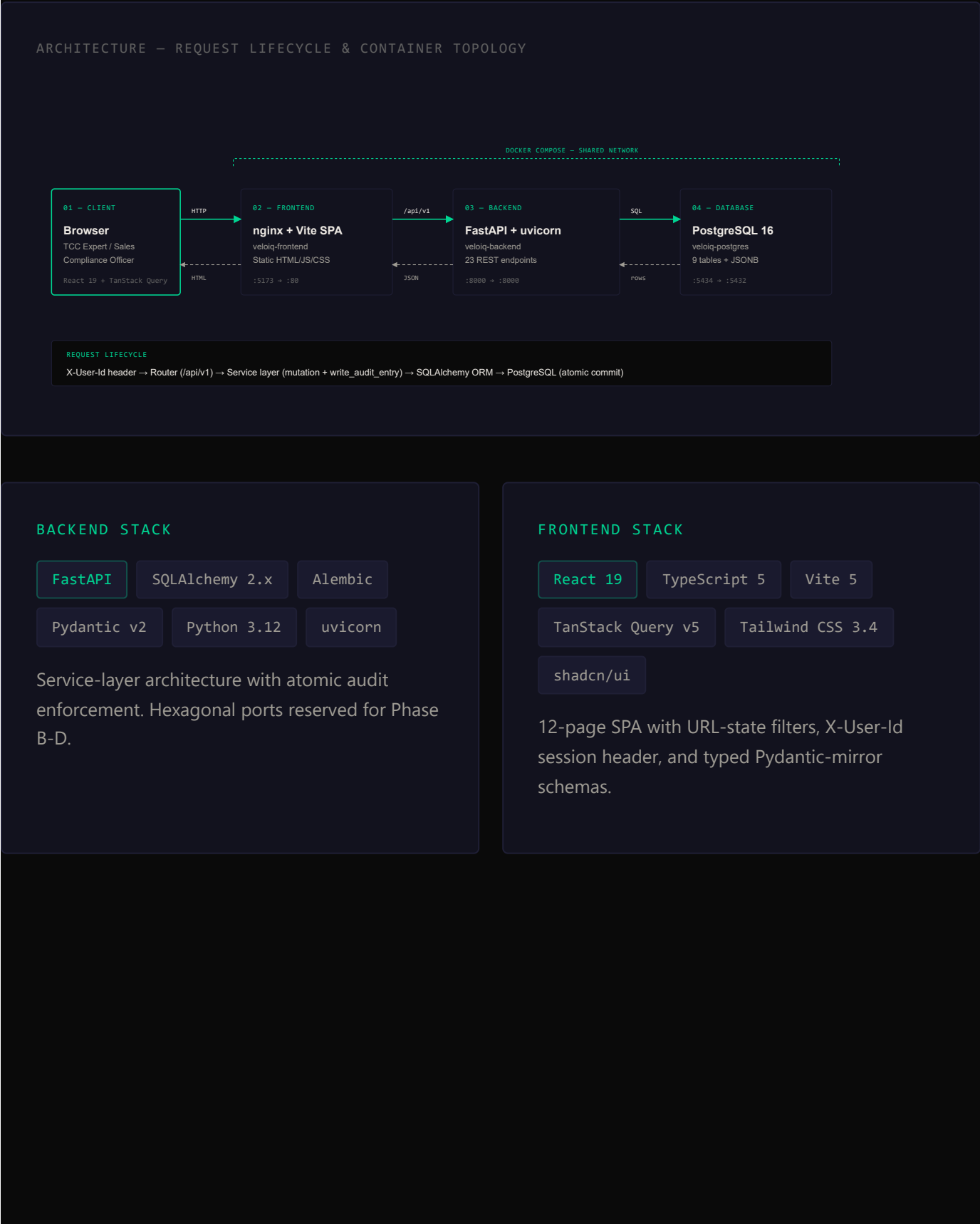


VELOCITY — CERTAINTY — TÜV RHEINLAND POC — V0.1.0 PHASE A

02 - SYSTEM ARCHITECTURE

Three-tier Docker topology

VeloIQ runs as three containerised services orchestrated through Docker Compose. A React SPA served behind nginx talks to a FastAPI backend, which owns all database writes through a service layer that enforces atomic audit logging.



Nine tables, five pipeline stages

Tables are colour-coded by their role in the 4-stage pipeline. The **audit_log** table is cross-cutting and append-only — the application role has UPDATE and DELETE revoked at the PostgreSQL layer, making audit entries immutable once committed.

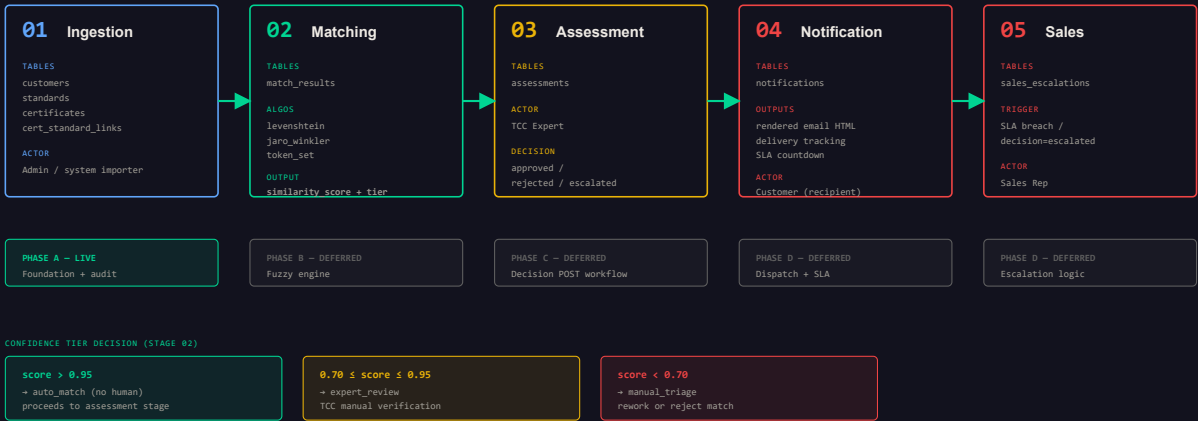
Migration 001 executes `REVOKE UPDATE, DELETE ON audit_log FROM veloiq` after table creation. The application database role **physically cannot mutate** audit entries after commit. This is the architectural cornerstone for TÜV non-repudiation — every mutation is provable post-hoc.

04 — DATA PIPELINE

Four stages — ingest, match, assess, notify

Each pipeline stage transforms inputs into auditable outputs. Phase A delivers the foundation tables + audit enforcement. Phases B, C, D add the matching engine, decision workflow, and notification dispatch respectively.

PIPELINE FLOW — DATA TRANSFORMATION & PHASE DELIVERY

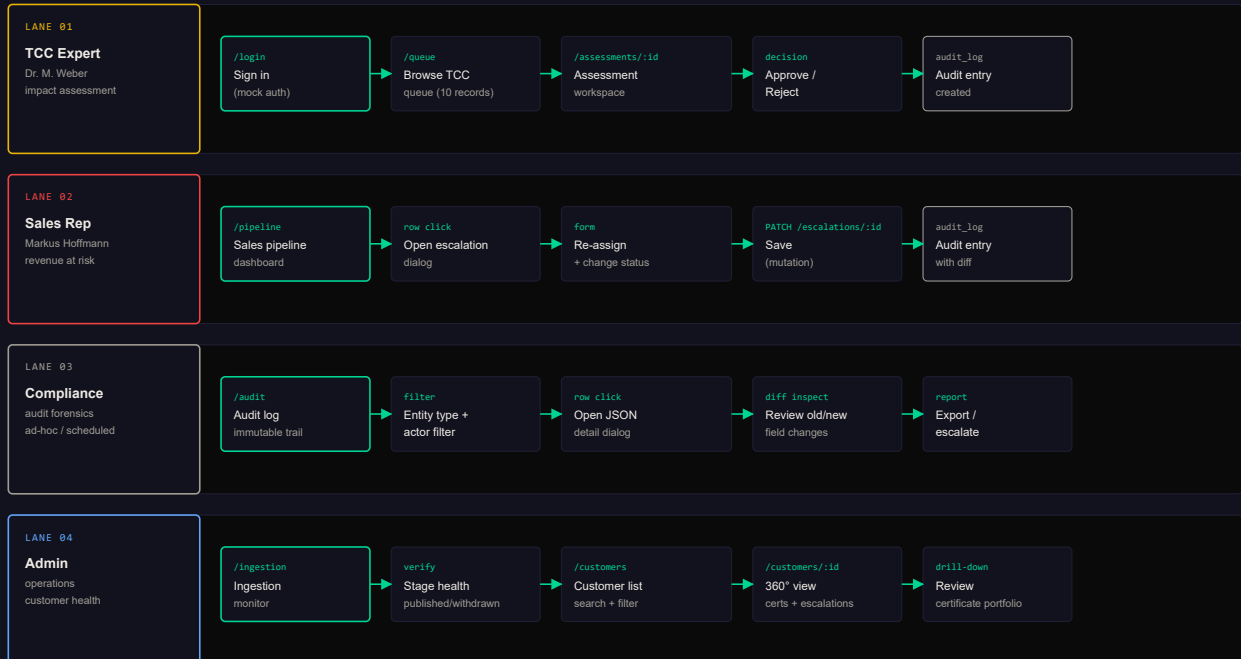


05 — USER JOURNEYS

Four personas, four flows

Each persona has a primary daily loop through VelolQ. The TCC Expert and Sales Rep are the most active; the Compliance Officer and Admin access the system on-demand.

USER JOURNEYS — SWIMLANE PER PERSONA



Twelve screens

VeloIQ exposes twelve routes across the four pipeline stages. Each screen is paired with its live API sources and the primary actor who depends on it.

Login

Mock-auth sign-in. Three seeded TCC expert identities are available as one-click demo buttons. The selected identity is stored in localStorage and injected as X-User-Id on every API request for audit attribution.

**VeloIQ**

STANDARDS AUTOMATION PLATFORM

Matching velocity. Compliance certainty.

TÜV Rheinland's automated pipeline for standards ingestion, fuzzy matching, expert assessment, and customer notification.

50
STANDARDS

200
CERTIFICATES

30
ASSESSMENTS

POC demo · v0.1.0 · Phase A Foundation

ACCESS

Sign in to VeloIQ

Enter your credentials or use a demo identity below.

EMAIL

you@tuv.example.com

PASSWORD

Sign in →

DEMO MODE

Sign in as a seeded TCC expert:

Dr. M. Weber

Dr. S. Chen

Dr. R. Kumar

GET /LOGIN

ACTOR: ALL USERS · DATA: NONE

http://localhost:8765/ARCHITECTURE_REPORT.html

6/26

System Dashboard

Pipeline-wide throughput KPIs with per-stage record counts. Each KPI card and stage row is a clickable link that deep-links into the matching list view.

 **VelolQ**

Dashboard

TCC Queue

Match Queue

Customers

Notifications

Sales Pipeline

Ingestion

Audit Log

 **Dr. M. Weber**
TCC Expert

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System Dashboard

06 Apr 2026

PIPELINE OVERVIEW

Real-time throughput and compliance state across the ingestion pipeline.

STANDARDS
INGESTED

10

MATCHES
GENERATED

10

33% auto-match

ASSESSMENTS
COMPLETE

10

NOTIFICATIONS
SENT

8

PIPELINE HEALTH

01	Ingestion	10	RECORDS
02	Fuzzy matching	10	RECORDS
03	Expert assessment	10	RECORDS
04	Customer notification	8	RECORDS

ESCALATIONS OPEN

3

Sales pipeline items

10 certificates tracked

- 1

KPI cards — click any to jump into that resource’s list view

2

Pipeline health panel shows records per stage (01–04)

3

Escalations open card deep-links to /pipeline

4

Certificates-tracked surfaces as a secondary indicator under escalations

GET /

ACTOR: ALL · DATA: 6 RESOURCE COUNTS (PAGINATION.TOTAL)

TCC Queue

Assessor work queue. Filterable by impact classification, decision, and assessor. Each row shows the assessor's decision, action required, reason code, decision timestamp, and a short cryptographic signature hash for integrity verification.

 **VeloIQ**

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DM

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TCC Queue

06 Apr 2026

WORK QUEUE

TCC Assessments

10 records

IMPACT

DECISION

ASSESSOR

All

All

Dr. M. Weber

ASSESSOR	IMPACT	Action	DECISION	REASON	DECIDED AT	SIG
Dr. A. Müller	major technical	Retest	ESCALATED	RC-MJ-60601-A	05 Apr 2026 · 03:06	c142f20c...
Dr. A. Müller	minor technical	Retest	APPROVED	RC-MN-62443-A	03 Apr 2026 · 03:06	b323bec4...
Dr. M. Weber	administrative	Reconfirm	APPROVED	RC-AD-9001-A	01 Apr 2026 · 03:06	271a7e68...
Dr. S. Chen	administrative	Reconfirm	APPROVED	RC-AD-2368-A	30 Mar 2026 · 03:06	45b125e6...
Dr. M. Weber	no change	Reconfirm	APPROVED	RC-NC-2015-B	29 Mar 2026 · 03:06	dd11c934...
Dr. M. Weber	no change	Reconfirm	APPROVED	RC-NC-2015-A	25 Mar 2026 · 03:06	048031c3...
Dr. R. Kumar	minor technical	Reconfirm	APPROVED	RC-MN-14001-B	19 Mar 2026 · 03:06	e3ff6024...
Dr. S. Chen	no change	Reconfirm	APPROVED	RC-NC-2016-C	12 Mar 2026 · 03:06	eb160c15...
Dr. L. Schmidt	no change	Reconfirm	REJECTED	RC-NC-14001-C	05 Mar 2026 · 03:06	52d3593f...
Dr. L. Schmidt	safety critical	Suspend	REJECTED	RC-SC-61010-A	25 Feb 2026 · 03:06	8d3272a4...

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Next >

1 Filter bar — Impact / Decision / Assessor (with autocomplete=off)

3 Reason code in RC-MAJOR-TECHNICAL format

2 Impact classification badge colour-coded (safety_critical in red)

4 Signature hash (first 8 chars) proves decision cryptographic integrity

GET /QUEUE

ACTOR: TCC EXPERT • DATA: /API/V1/ASSESSMENTS

Assessment Workspace

Detailed assessment view with clause delta comparison (year-over-year diff), impact summary metrics, and a decision form. In Phase A the Approve/Reject buttons show a toast only; in Phase C they POST a persistent decision.

VelolQ

Dashboard

TCC Queue

Match Queue

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Sales Pipeline

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Audit Log

Assessment Workspace

06 Apr 2026

← TCC Queue > IEC 60601-1:2020

ASSESSMENT WORKSPACE

Medical electrical equipment — Part 1: General requirements for basic safety and essential performance

Assessment ID: 33bfa7e2-b017-4ed4-b031-8f8201174ef1

0.819 · EXPERT REVIEW

ESCALATED

01 — CLAUSE DELTA (2018 — 2023)

§ 4.2.1

Annex A — Hazard classification

2018: Category I/II/III classification based on pressure × volume.

2023: Extended to include fluid group classification (Group 1/2) with revised thresholds at 50 bar.

MODIFIED

§ 6.3.2

Test procedure — Pressure cycling

2018: Minimum 1,000 pressure cycles at 1.5× working pressure.

2023: Increased to 2,500 pressure cycles at 1.43× working pressure, with intermediate visual inspection at cycle 1,000.

MODIFIED

§ 7.1.4

Documentation — CE technical file

2023: New requirement: technical file must include digital twin simulation results for all variants (> 100 L capacity).

ADDED

02 — IMPACT SUMMARY

AFFECTED CERTIFICATES

8

AFFECTED CUSTOMERS

3

CLAUSE DELTAS

3

PRIORITY

HIGH

03 — RECORD DECISION

IMPACT CLASSIFICATION *

Major technical

ACTION REQUIRED *

Retest

REASON CODE *

RC-MJ-60601-A

NOTES

IEC 60601-1 Ed.3.2:2020 introduces new essential performance requirements (Clause 4.3) and requires updated risk management file per ISO 14971:2019. Rolls-Royce EECU medical electrical cert must retest — full Ed.3.2 type test campaign required.

Approve

Reject

Decided by Dr. A. Müller · 05 Apr 2026

POC demo mode: Approve/Reject show toast confirmation but do NOT persist to the backend. Full POST workflow is Phase C.

DM

Dr. M. Weber

TCC Expert

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1

Clause delta panel shows 2018 → 2023 diffs with MODIFIED/ADDED tags

2

Impact summary KPIs: affected certs, customers, deltas, priority

3 Decision form: impact classification, action, reason code, notes

4 Approve / Reject buttons (Phase A: toast only; Phase C: persistent)

GET /ASSESSMENTS/:ID

ACTOR: TCC EXPERT · DATA: /API/V1/ASSESSMENTS/:ID

Customer Directory

Paginated customer table sorted by country then company name (client-side secondary sort). Country and sales area filters are URL-backed. Click any row to open the Customer 360 view.

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Dashboard

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Match Queue

Customers

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Ingestion

Audit Log

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06 Apr 2026

CUSTOMER DIRECTORY

Customers 18 records

COUNTRY

SALES AREA

All

All

CUSTOMER #	COMPANY	COUNTRY	SALES AREA	LANG	CONTACT
CUST-0006	Hitachi Energy Ltd.	CN	Greater China	EN	Chen Hao
CUST-0005	Huawei Technologies Co., Ltd.	CN	Greater China	ZH	Li Wei
CUST-0003	ABB Automation AG	DE	EMEA	DE	Stefan Weiss
CUST-0002	Bosch Rexroth GmbH	DE	EMEA	DE	Ingrid Müller
CUST-0004	Schneider Electric SE	DE	EMEA	EN	Marie Laurent
CUST-0001	Siemens AG	DE	EMEA	DE	Demo Contact
CUST-0010	Rolls-Royce Holdings plc	GB	EMEA	EN	James Whitfield
CUST-0009	Mahindra & Mahindra Ltd.	IN	South Asia	EN	Rajesh Patel
CUST-0008	Tata Steel Limited	IN	South Asia	EN	Priya Sharma
CUST-0007	Honeywell International Inc.	US	Americas	EN	Sarah Johnson

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GET /CUSTOMERS

ACTOR: ADMIN · DATA: /API/V1/CUSTOMERS (10 SEED RECORDS)

Customer 360°

Single-company deep-dive: identity card, portfolio KPIs (active certs / expiring / open escalations / active notifications), and full certificate inventory.

 **VelolQ**

Dashboard

TCC Queue

Match Queue

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VelolQ06 Apr 2026

[← Customers](#) > CUST-0002

CUSTOMER 360

**Bosch Rexroth GmbH**
CUST-0002

COUNTRY
DE

SALES AREA
EMEA

LANGUAGE
DE

CONTACT

Ingrid Müller
ingrid.mueller@bos...

PORTFOLIO SUMMARY

ACTIVE CERTIFICATES
0

EXPIRING SOON
1

OPEN ESCALATIONS
0

ACTIVE NOTIFICATIONS
0

CERTIFICATE PORTFOLIO

CERT #	PRODUCT	STATUS	ISSUED	EXPIRES
TC-00003	Hydraulic Pressure Transducer HM20	EXPIRING	20 Jun 2023	04 Jun 2026

1 Header: company name + customer number (CUST-NNNN)

2 Identity panel: country, sales area, language, primary contact

3 Portfolio KPIs aggregate data from 4 different endpoints

4 Certificate portfolio table with status badges

GET /CUSTOMERS/:ID

ACTOR: ADMIN / SALES · DATA: 4 ENDPOINTS JOINED

http://localhost:8765/ARCHITECTURE_REPORT.html

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Match Queue

Fuzzy-match triage view showing the confidence distribution (auto_match / expert_review / manual_triage / pending). Algorithm scores for each candidate pair are shown side-by-side for transparency.

 **VelolQ**

Dashboard

TCC Queue

Match Queue

Customers

Notifications

Sales Pipeline

Ingestion

Audit Log

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TCC Expert

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VelolQ

06 Apr 2026

FUZZY MATCH TRIAGE

Match Queue 10 records

CONFIDENCE DISTRIBUTION

AUTO MATCH ↶
5

EXPERT REVIEW
3

MANUAL TRIAGE
2

PENDING
1

Metrics reflect current page (10 records) · 10 total

CONFIDENCE TIER

STATUS

All ▾

All ▾

COMPOSITE	LEVENSHTEIN	JARO - WINKLER	TOKEN SET	STATUS	MATCHED
0.810 · EXPERT REVIEW	0.755	0.832	0.810	PENDING	05 Apr 2026
0.998 · AUTO MATCH	0.995	0.999	1.000	REVIEWED	31 Mar 2026
0.996 · AUTO MATCH	0.993	0.998	1.000	REVIEWED	31 Mar 2026
0.972 · AUTO MATCH	0.940	0.985	0.960	REVIEWED	31 Mar 2026
0.958 · AUTO MATCH	0.920	0.972	0.945	REVIEWED	31 Mar 2026
0.892 · EXPERT REVIEW	0.855	0.910	0.880	REVIEWED	31 Mar 2026
0.842 · EXPERT REVIEW	0.780	0.870	0.830	REVIEWED	31 Mar 2026
0.680 · MANUAL TRIAGE	0.620	0.705	0.680	REVIEWED	31 Mar 2026
0.998 · AUTO MATCH	0.995	0.999	1.000	REVIEWED	31 Mar 2026
0.540 · MANUAL TRIAGE	0.480	0.560	0.590	REVIEWED	31 Mar 2026

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http://localhost:8765/ARCHITECTURE_REPORT.html

14/26

1 Confidence distribution KPIs across 3 tiers + pending

2 Composite similarity pill colour-coded by tier

3 4 raw algorithm scores: levenshtein, jaro-winkler,
token set

4 Status badge: pending (yellow) / reviewed (accent)

GET /MATCHES

ACTOR: TCC EXPERT • DATA: /API/V1/MATCHES

Match Forensic

Per-match trace view showing how raw strings were normalised, the four fuzzy-algorithm scores individually, and provenance (matched_at / reviewed_at / tier).

VelolQ

Dashboard

TCC Queue

Match Queue

Customers

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Sales Pipeline

Ingestion

Audit Log

Match Forensic View

06 Apr 2026

← Back > Match e3820d65...

MATCH FORENSIC VIEW

Medical electrical equipment — Part 1: General requirements for basic safety and essential performance

Match ID: e3820d65-d294-475d-a134-49a732500560

0.810 · EXPERT REVIEW

PENDING

01 — NORMALIZATION TRACE

NATOS SOURCE

RAW AC_CODE

IEC 60601-1:2020

→ lowercase + trim whitespace

NORMALIZED CODE

iec 60601-1:2020

SAP TARGET

RAW STANDARD_REF

DIN EN IEC 60601-1:2020

→ strip prefixes + lowercase + trim

NORMALIZED REF

iec 60601-1:2020

02 — ALGORITHM SCORES

COMPOSITE (SIMILARITY)

0.810

Weighted ensemble of all algorithms below

LEVENSHTEIN

0.755

Edit-distance-based; good for character-level matches

JARO-WINKLER

0.832

Weights prefix matches higher; good for standards

TOKEN SET

0.810

Set-based on tokens; order-invariant

03 — PROVENANCE

MATCHED AT

05 Apr 2026 · 03:06

REVIEWED AT

—

CONFIDENCE TIER

EXPERT REVIEW

Dr. M. Weber

TCC Expert

Sign out

← Back to queue

1

Normalization trace: raw NATOS ↔ raw SAP inputs with transforms

2

Algorithm scores: composite, levenshtein, jaro-winkler, token set

3

Each score annotated with what it measures + why it matters

4

Provenance band: matched_at, reviewed_at, confidence tier

GET /MATCHES/:ID

ACTOR: TCC EXPERT · DATA: /API/V1/MATCHES/:ID + /STANDARDS/:ID

http://localhost:8765/ARCHITECTURE_REPORT.html

16/26

Notification Center

Customer outreach delivery funnel: sent → delivered → opened → clicked, plus bounced / breached tallies. Each notification row shows subject, customer, language, status, and SLA countdown.

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Dashboard

TCC Queue

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VelolQ

06 Apr 2026

CUSTOMER OUTREACH

Notification Center 8 records

DELIVERY FUNNEL

SENT 

DELIVERED

OPENED

CLICKED

BOUNCED

BREACHED

0

2

2

2

0

2

Funnel reflects current page (8 records) · 8 total

STATUS

CUSTOMER ID (UUID)

All

Paste customer UUID

SUBJECT	CUSTOMER	LANG	STATUS	SENT
TÜV Rheinland — ISO 13485:2016: confirmation required	9b3d96b8...	EN	CLICKED	21 Jul 2026
TÜV Rheinland — ISO 14001:2015: confirmation required	96344b91...	EN	CLICKED	21 Jun 2026
TÜV Rheinland — ISO 45001:2018: confirmation required	f6b9f165...	EN	DELIVERED	21 Jun 2026
TÜV Rheinland — ISO 45001:2018: confirmation required	ca3f208f...	EN	OPENED	11 Jun 2026
TÜV Rheinland — ISO 9001:2015: confirmation required	69b8c625...	DE	DELIVERED	22 May 2026
TÜV Rheinland — IEC 62368-1:2023: confirmation required	24c66b9f...	ZH	OPENED	07 May 2026
TÜV Rheinland — IEC 62443-3-3:2023: retest required	69b8c625...	DE	BREACHED	18 Mar 2026
TÜV Rheinland — IEC 60601-1:2020: retest required	9b3d96b8...	EN	BREACHED	11 Mar 2026

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TCC Expert

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1

Delivery funnel: 6 KPI cards for each notification status

2

Status filter + customer UUID filter

3

Template language column (EN / DE / ZH)

4

Status badge: breached (red) signals SLA violation

GET /NOTIFICATIONS

ACTOR: TCC EXPERT • DATA: /API/V1/NOTIFICATIONS

Sales Pipeline

Revenue-at-risk dashboard: open / contacted / resolved counts plus total opportunity value. Each escalation row has a Manage button opening a PATCH mutation dialog for assignee + status.

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06 Apr 2026

SALES REVENUE AT RISK

Sales Pipeline

3 records

PIPELINE SUMMARY

OPEN

1

CONTACTED

2

RESOLVED

0

TOTAL OPPORTUNITY

\$

\$660,000

Metrics reflect current page (3 records) • 3 total

STATUS

ASSIGNEE

CUSTOMER ID (UUID)

All

Assignee user ID

Paste customer UUID

REASON	CUSTOMER	OPPORTUNITY	ASSIGNED TO	STATUS	CREATED	
sla breach	f6b9f165...	\$125,000	Dr. M. Weber	CONTACTED	05 Jul 2026	Manage
retest overdue	69b0c625...	\$85,000	Markus Hoffmann	OPEN	01 Apr 2026	Manage
major change retest	9b3d96b8...	\$450,000	Jennifer Walsh	CONTACTED	25 Mar 2026	Manage

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GET /PIPELINE

ACTOR: SALES REP • DATA: /API/V1/ESCALATIONS (+ PATCH)

Manage Escalation (dialog)

Modal dialog with PATCH mutation form. Saving fires PATCH /escalations/:id which writes a paired audit entry with old/new field diff.

VelolQ

SALES REVENUE AT RISK

Dashboard

TCC Queue

Match Queue

Customers

Notifications

Sales Pipeline

Ingestion

Audit Log

SALES REVENUE AT RISK

Sales Pipeline 3 records

PIPELINE SUMMARY

OPEN 1

CONTACTED 2

RESOLVED 0

TOTAL OPPORTUNITY \$660,000

retest overdue 69b0c625... \$85,000 Markus Hoffmann OPEN 01 Apr 2026

major change retest 963d96b9... \$450,000 Jennifer Walsh CONTACTED 25 Mar 2026

DM Dr. M. Weber TCC Expert

[> Sign out]

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1

Escalation reason + entity ID shown in header

2

Status dropdown: open / contacted / resolved

3

Assignee text input (prefilled from current assignee)

4

Save fires PATCH mutation + creates audit_log entry

PATCH /ESCALATIONS/:ID

ACTOR: SALES REP • MUTATION: STATUS, ASSIGNED_TO

http://localhost:8765/ARCHITECTURE_REPORT.html

19/26

Ingestion Monitor

Standards catalog with ISO stage codes: Published (60) / Draft (10) / Withdrawn (90). Superseded counts derive from non-null replaced_by. Filters on stage, committee, and base number.

 **VelolQ**

Dashboard

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VelolQ

06 Apr 2026

STANDARDS CATALOG

Ingestion Monitor 10 records

CATALOG SUMMARY

PUBLISHED

8

SUPERSEDED

1

WITHDRAWN

1

LATEST INGESTED

06 Apr 2026

Metrics reflect current page (10 records) · 10 total

STAGE

COMMITTEE

BASE NUMBER

All

e.g. ISO/TC 176

e.g. 9001

AC CODE	TITLE	STAGE
ISO 9001:2015	Quality management systems — Requirements	Published
ISO 14001:2015	Environmental management systems — Requirements with gu...	Published
ISO 45001:2018	Occupational health and safety management systems — Req...	Published
IEC 62368-1:2023	Audio/video, information and communication technology equ...	Published
IEC 61010-1:2010	Safety requirements for electrical equipment for measureme...	Published
ISO 13485:2016	Medical devices — Quality management systems — Requirem...	Published
IEC 60601-1:2020	Medical electrical equipment — Part 1: General requirements f...	Withdrawn
IEC 60601-1:2024	Medical electrical equipment — Part 1: General requirements f...	Draft
ISO 27001:2022	Information security, cybersecurity and privacy protection — ...	Published
IEC 62443-3-3:2023	Industrial communication networks — Network and system se...	Published

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- ACTOR: ADMIN · DATA: /API/V1/STANDARDS

ACTOR: COMPLIANCE OFFICER · DATA: /API/V1/AUDIT

Audit Entry (detail dialog)

JSON detail view showing the full audit entry including old/new field diffs, escalation reason, actor, and IP address. Rendered via monospace <pre> with syntax-preserved indentation.

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Ingestion

Audit Log

IMMUTABLE AUDIT TRAIL

Audit entry · updated

sales_escalation · 2b416f98-99e4-4aee-8b86-885379941aad

ACTOR	CREATED AT	IP ADDRESS
Dr. M. Weber	06 Apr 2026 03:34	-

```
{
  "changed_fields": {
    "status": {
      "new": "contacted",
      "old": "resolved"
    },
    "assigned_to": {
      "new": "Dr. M. Weber",
      "old": "Priya Kapoor"
    }
  },
  "escalation_reason": "sla_breach"
}
```

Dr. M. Weber

ICC Expert

Sign out

1

Dialog header: entity_type + entity_id (full UUID)

2

Metadata strip: actor, created_at, ip_address

3

Full JSON diff: old/new values per changed_fields

4

Escalation reason preserved alongside diff for context

GET /AUDIT/:ID ACTOR: COMPLIANCE OFFICER · DATA: /API/V1/AUDIT/:ID

http://localhost:8765/ARCHITECTURE_REPORT.html

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23 REST endpoints

All endpoints live under `/api/v1`. Mutations (POST/PATCH) require the `X-User-Id` header. Reads are open. Every mutation writes a paired `audit_log` entry in the same transaction.

METHOD	PATH	PURPOSE	TYPE
GET	/standards	List standards (status, committee, base_number filters)	READ
GET	/standards/{id}	Retrieve single standard	READ
POST	/standards	Ingest new standard + audit entry	MUT
PATCH	/standards/{id}	Update standard fields + audit diff	MUT
GET	/customers	List customers (country, sales_area)	READ
GET	/customers/{id}	Retrieve single customer	READ
POST	/customers	Create customer + audit	MUT
PATCH	/customers/{id}	Update contact info + audit diff	MUT
GET	/certificates	List certs (customer_id, status)	READ
GET	/certificates/{id}	Retrieve single certificate	READ
POST	/certificates	Create certificate + audit	MUT
PATCH	/certificates/{id}	Update status/dates + audit diff	MUT
GET	/matches	List matches (confidence_tier, status)	READ
GET	/matches/{id}	Retrieve match with algo scores	READ
GET	/assessments	List assessments (assessor_id, decision, impact)	READ
GET	/assessments/{id}	Retrieve single assessment	READ
GET	/notifications	List notifications (customer_id, status)	READ

METHOD	PATH	PURPOSE	TYPE
GET	/notifications/{id}	Retrieve single notification	READ
GET	/escalations	List escalations (status, customer_id, assigned_to)	READ
GET	/escalations/{id}	Retrieve single escalation	READ
PATCH	/escalations/{id}	Update status/assignee + audit diff	MUT
GET	/audit	List audit entries (filters on entity, actor, action, date)	READ
GET	/audit/{id}	Retrieve single audit entry	READ

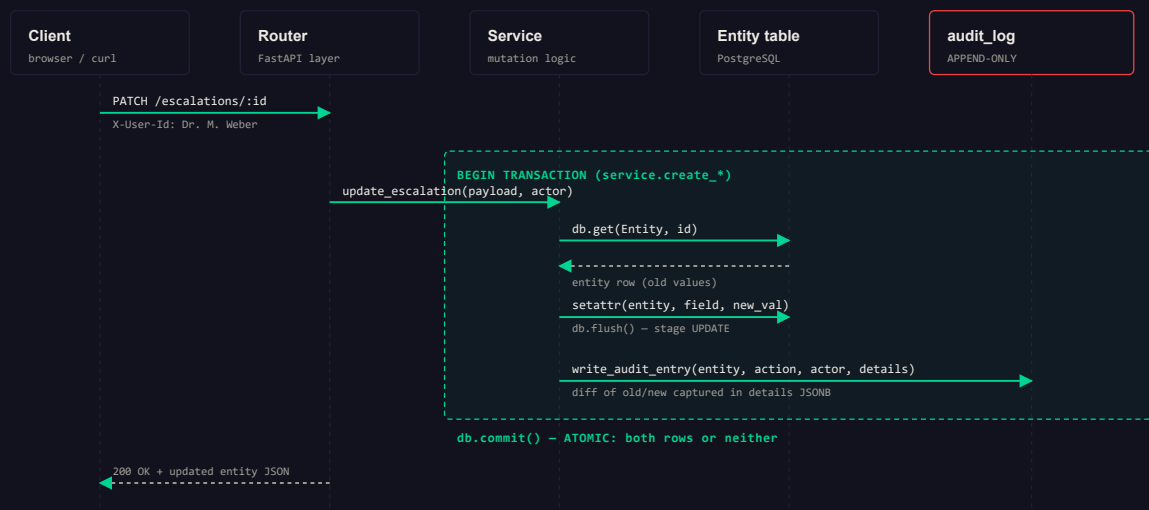
Totals: 23 endpoints · 8 routers · 9 mutations (4 POST, 5 PATCH) · 14 reads. No DELETE endpoints exposed — data is never removed, only superseded via status updates.

08 — AUDIT ENFORCEMENT

One transaction, two rows

Every mutation writes two rows atomically: the entity change and its audit entry. If the audit write fails, the entity change rolls back. The application database role cannot UPDATE or DELETE audit entries post-commit.

AUDIT SEQUENCE — ATOMIC MUTATION + AUDIT WRITE



Reference implementation

Extract from backend/app/services/certificates_service.py:24-59 — the canonical pattern every mutation service follows:

```

def create_certificate(
    db: Session,
    payload: CertificateCreate,
    actor: str,
) -> Certificate:
    """Create a Certificate and write a paired audit_log entry."""
    cert = Certificate(
        id=uuid4(),
        certificate_number=payload.certificate_number,
        customer_id=payload.customer_id,
        product_description=payload.product_description,
        status=payload.status,
        issue_date=payload.issue_date,
        expiry_date=payload.expiry_date,
    )
    db.add(cert)
    db.flush()
  
```

```
write_audit_entry(  
    db,  
    entity_type="certificate",  
    entity_id=cert.id,  
    action="created",  
    actor=actor,  
    details={  
        "certificate_number": cert.certificate_number,  
        "customer_id": str(cert.customer_id),  
        "status": cert.status,  
        "issue_date": cert.issue_date.isoformat(),  
        "expiry_date": cert.expiry_date.isoformat(),  
    },  
)
```

```
db.commit()  
db.refresh(cert)  
return cert
```

ATOMICITY

Both INSERT (entity) and INSERT (audit_log) execute in the same `db.commit()`. If either fails, the transaction rolls back and no rows are written. The pairing is guaranteed at the database level, not just at application level.

IMMUTABILITY

Migration 001 executes `REVOKE UPDATE, DELETE ON audit_log FROM veloiq`. The application role owns audit entries only in their create phase. Once committed, no SQL executed by the application can alter them.